



Tool Life and Machining Economics

Groover M.P.: Chapters 19 and 20



The student:

- understands the mechanism of cutting tool wear;
- knows materials for cutting tools;
- knows about machining economics.





THREE MODES OF TOOL FAILURE

1. Fracture failure

- Cutting forces and/or dynamic vibrations become excessive, leading to brittle fracture

2. Temperature failure

- Cutting temperature is too high for the tool material

3. Gradual wear

- Gradual wearing of the cutting tool



Phenomena causing tool wear are of the following nature:

- **Mechanical:** mainly due to the abrasion of the chip on the cutting tool rake face;
- **Thermal:** during cutting temperatures rise noticeably;
- **Chemical:** with the temperature increase, we can observe oxidation or solid state diffusion mechanisms that reduce mechanical characteristics of the tool.

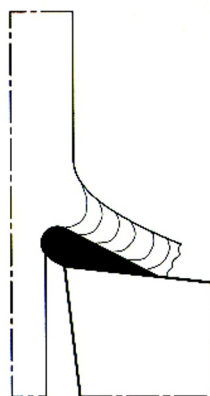


Possible **mechanisms** of wear formation:

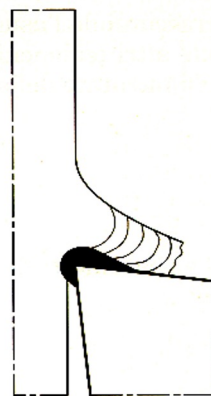
- **Abrasion:** the tool meets high hardness particles;
- **Diffusion:** atoms passage between part and tool materials;
- **Oxidation:** oxygen from the atmosphere forms easily removable oxides;
- **Adhesion:** phenomenon of the built-up edge;
- **Plastic deformation:** softening due to mechanical and thermal effects;
- **Fatigue:** due to mechanical and thermal cycles.



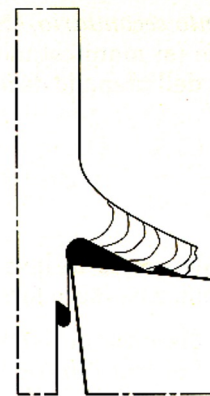
Built-up edge



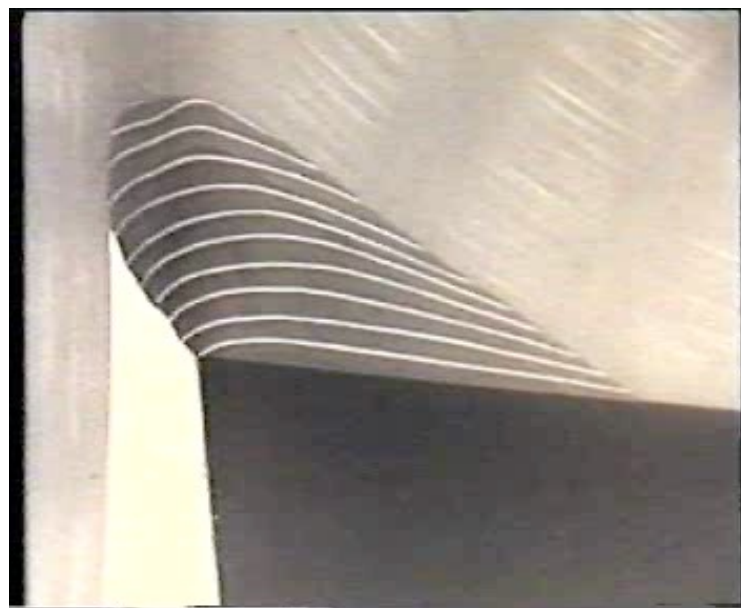
a)



b)



c)



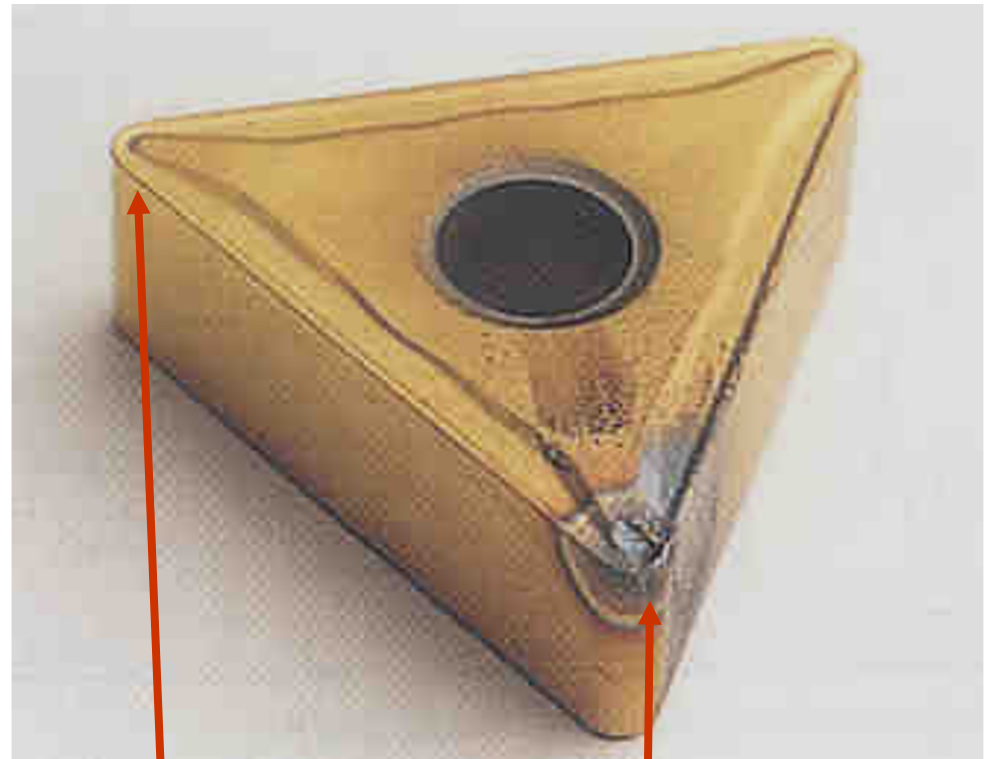
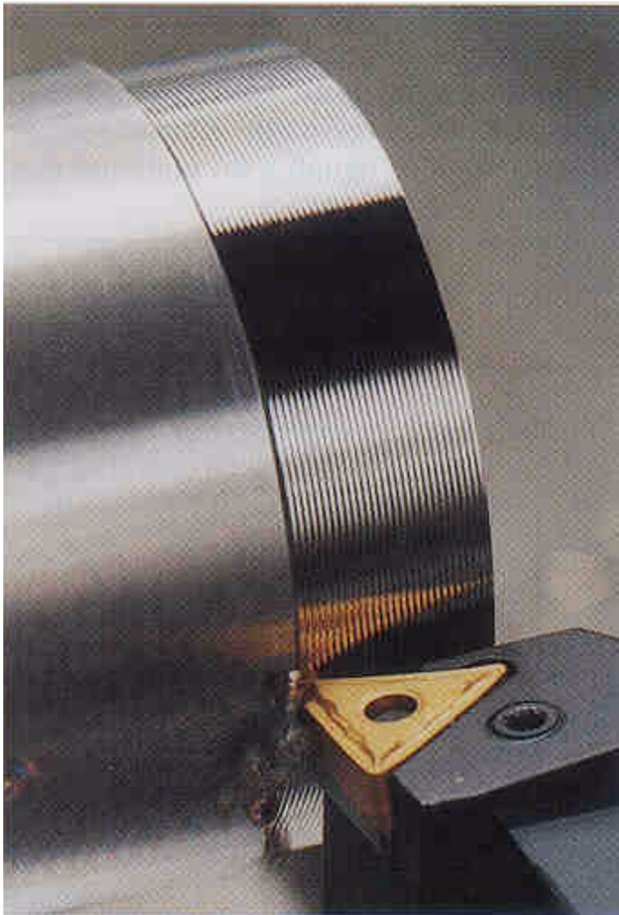


Relevant factors influencing the tool wear:

- Tool characteristics;
- Temperature of the cutting zone (cutting speed);
- Cutting parameters;
- Cooling of the cutting zone;
- Presence of thermal cycles;
- Lubrication of the cutting zone;
- Chemical affinity between the tool and the work material;
- ...



During machining tool ***gradual wear occurs***.
Gradual wear is the ***preferred mode of tool failure***, because it leads to the longest possible use of the tool.

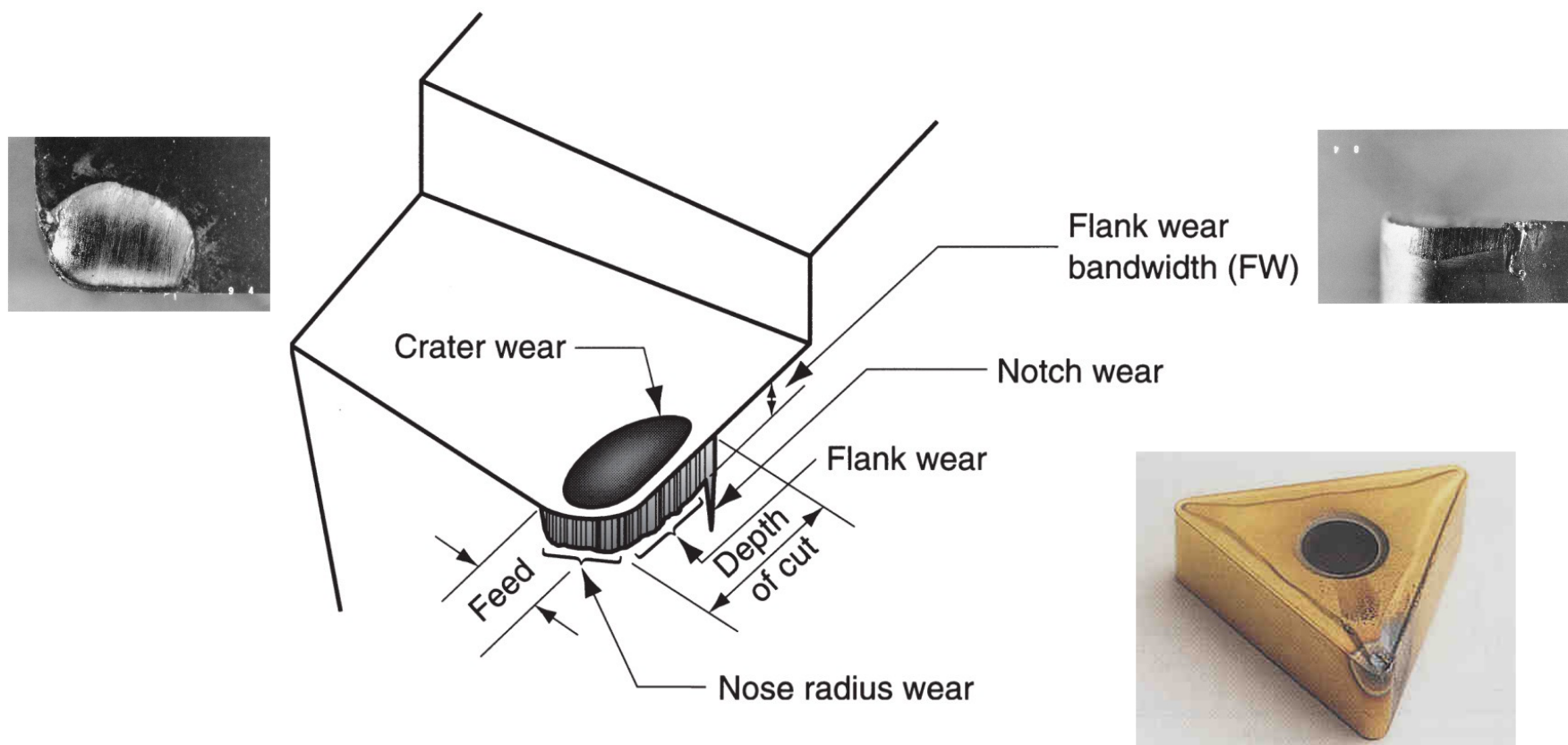


New cutting
edge

Worn cutting
edge



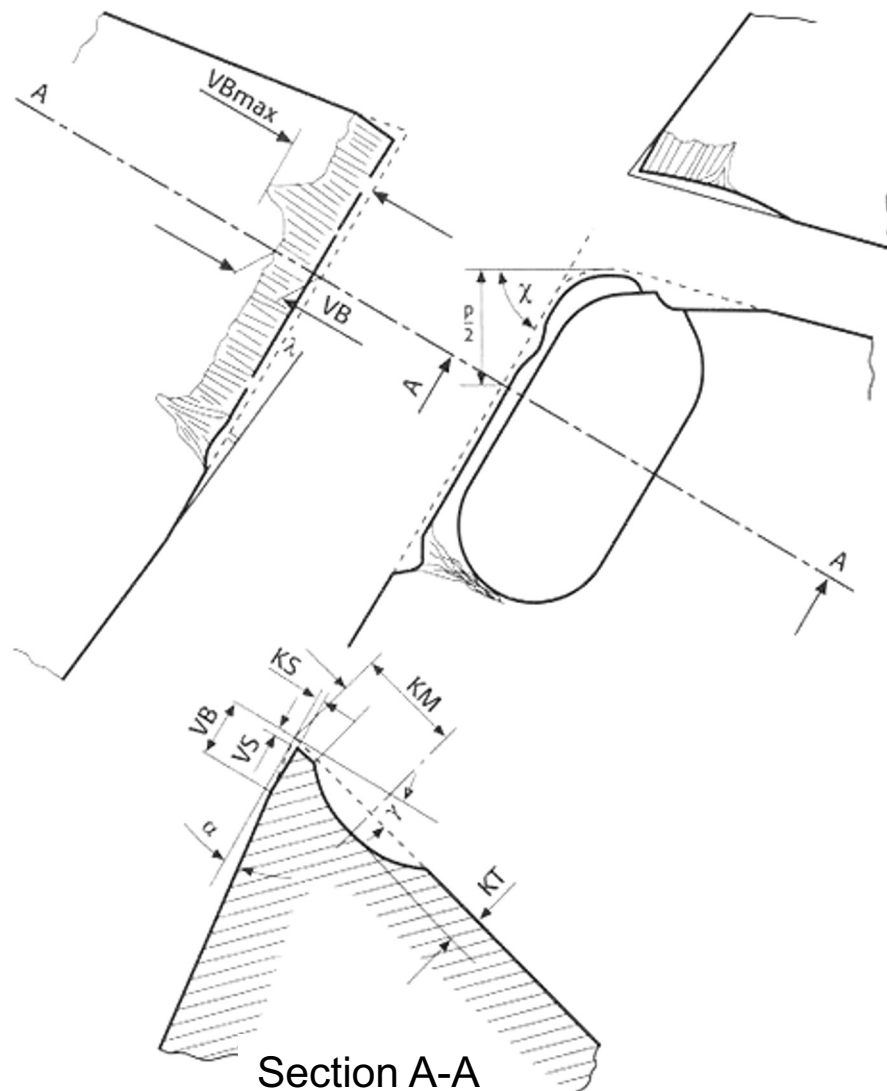
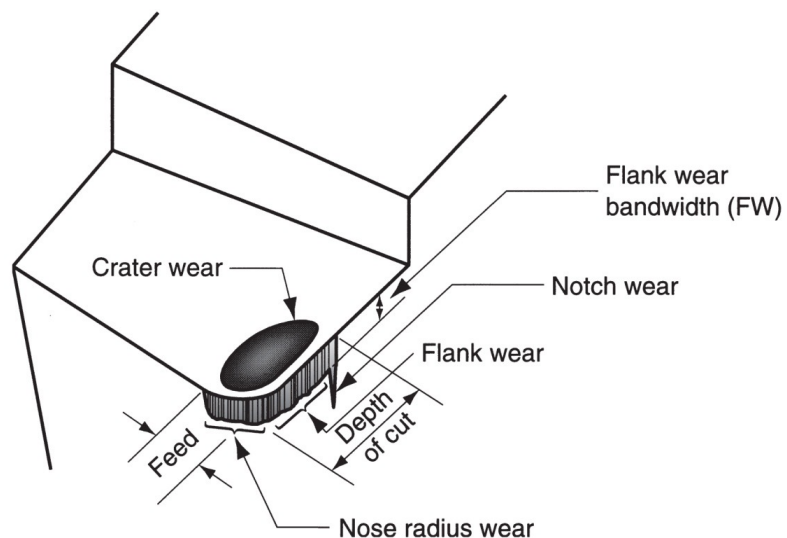
Worn cutting tool, showing the principal locations and types of wear that occur.





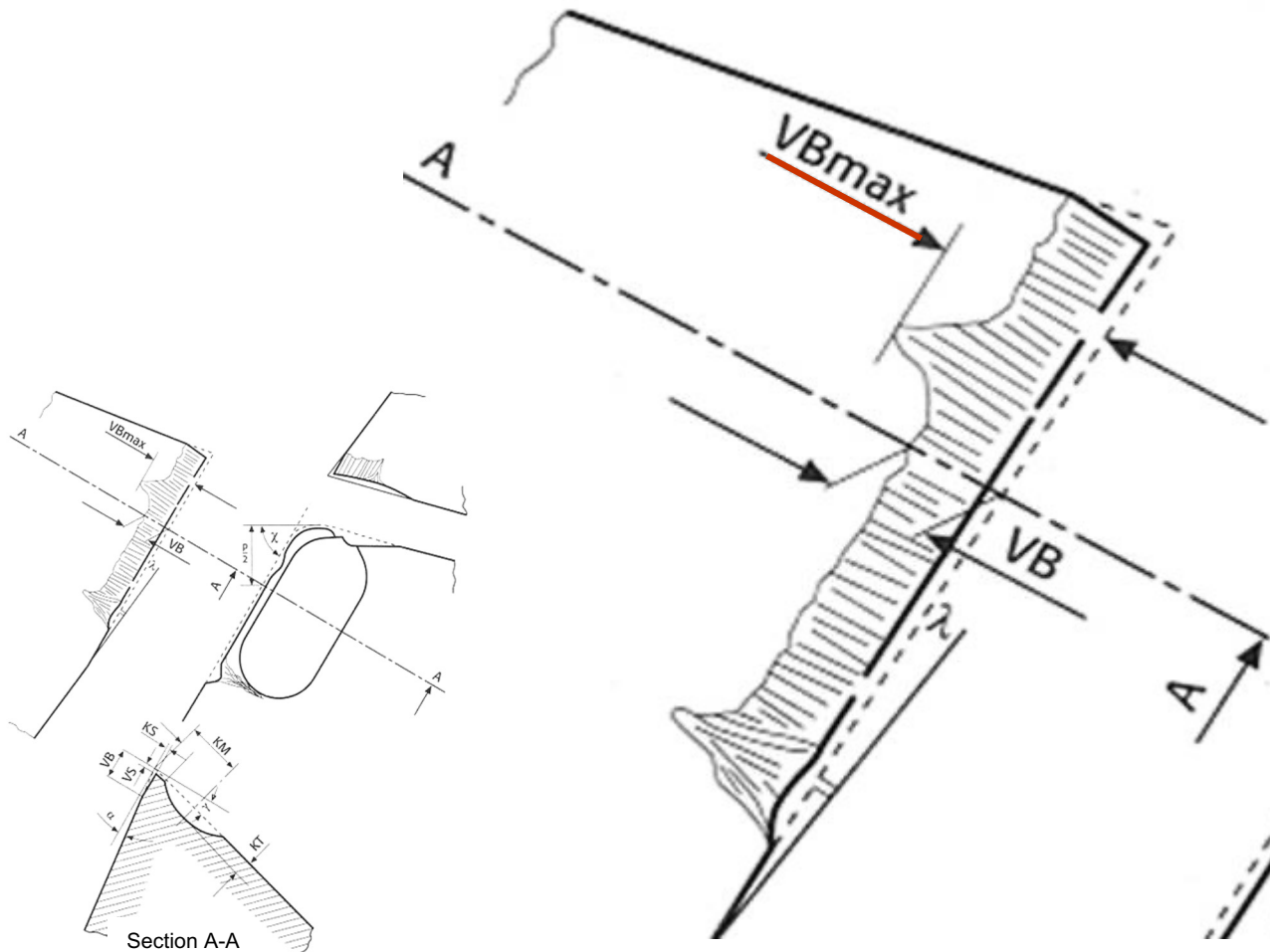
Gradual wear occurs at two locations on a tool:

- **Crater wear:** occurs on top rake face.
- **Flank wear:** occurs on flank (side of tool).





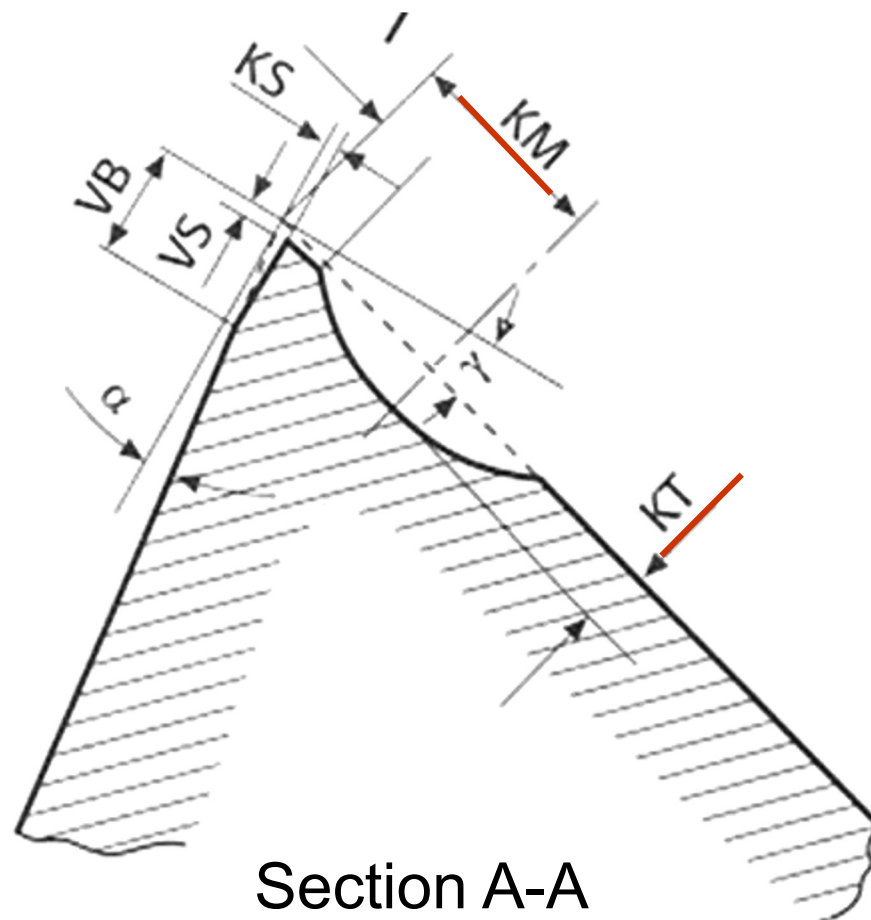
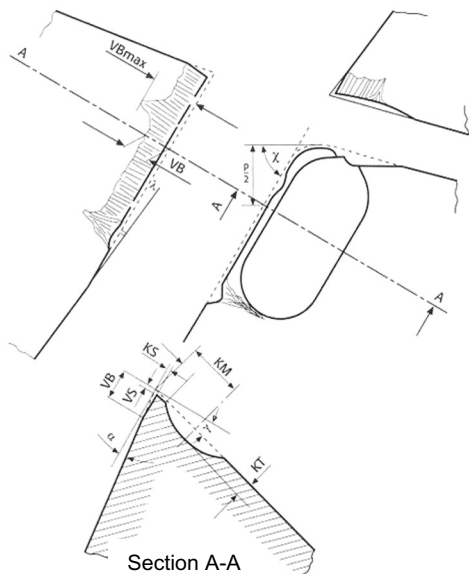
Flank wear affects dimensional accuracy and surface finish. It is quantified mainly by the parameter **VB** (average VB and VB_{max}).





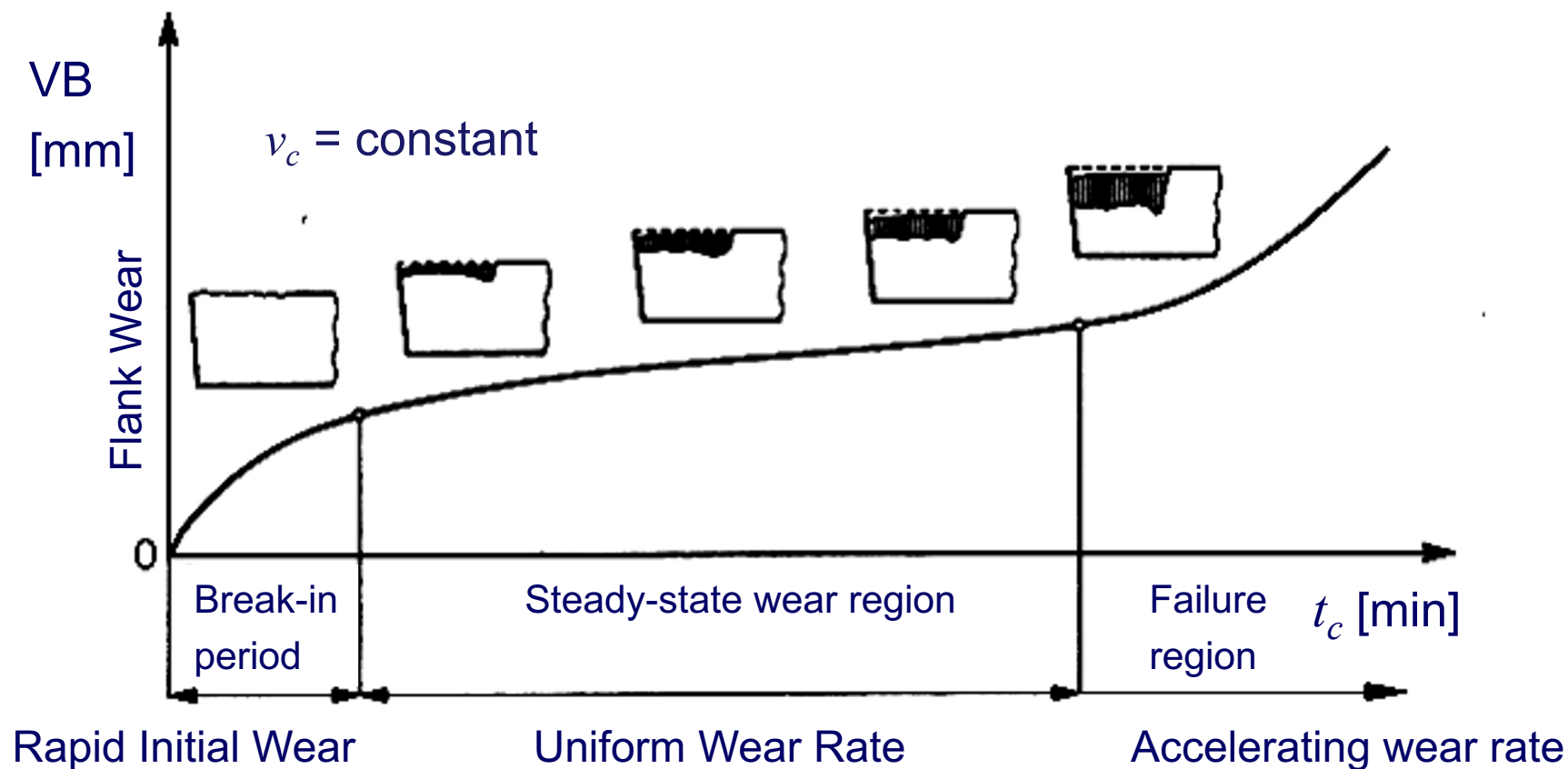
Crater wear weakens the tool. It is quantified by:

- **KT** : crater depth;
- **KM** : position of the deepest point of the crater measured from the original cutting edge.



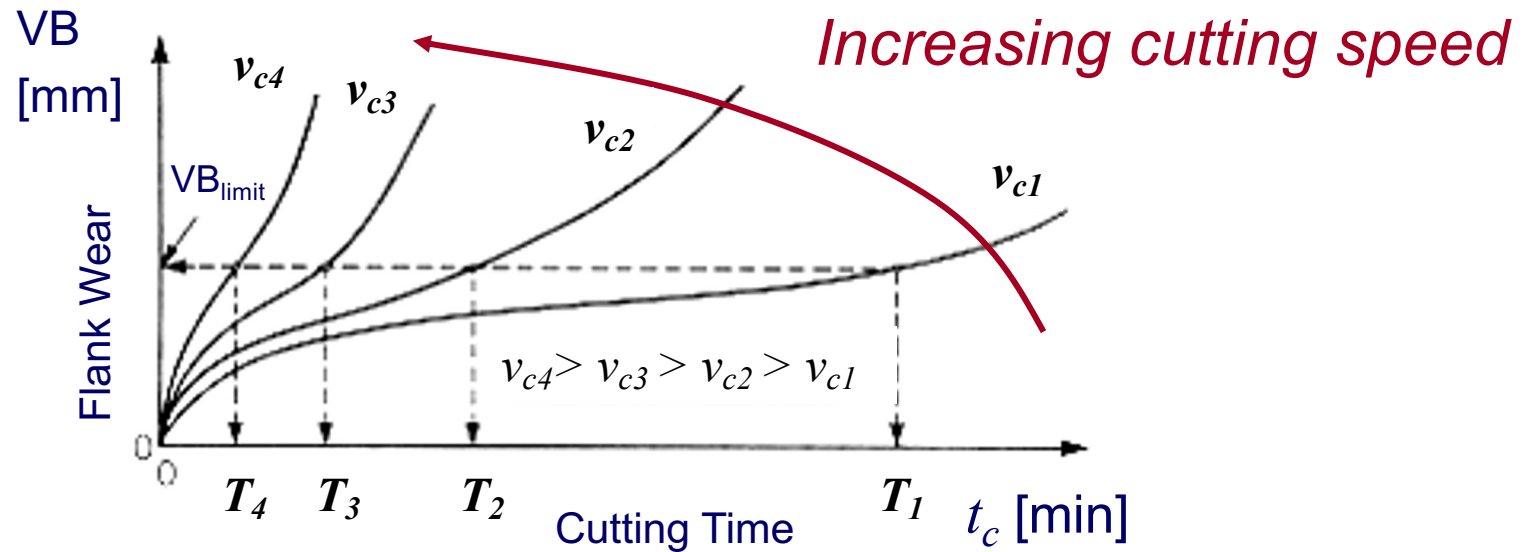


Tool wear as a function of cutting time.



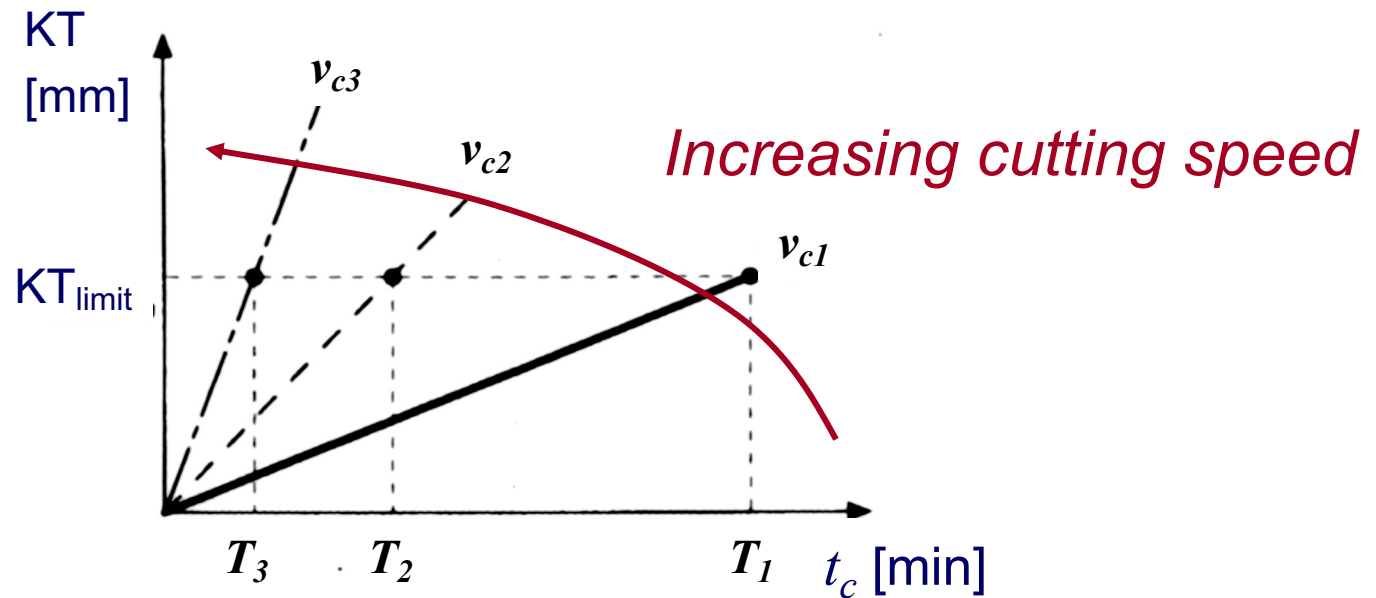


Effect of cutting speed on flank wear (VB)





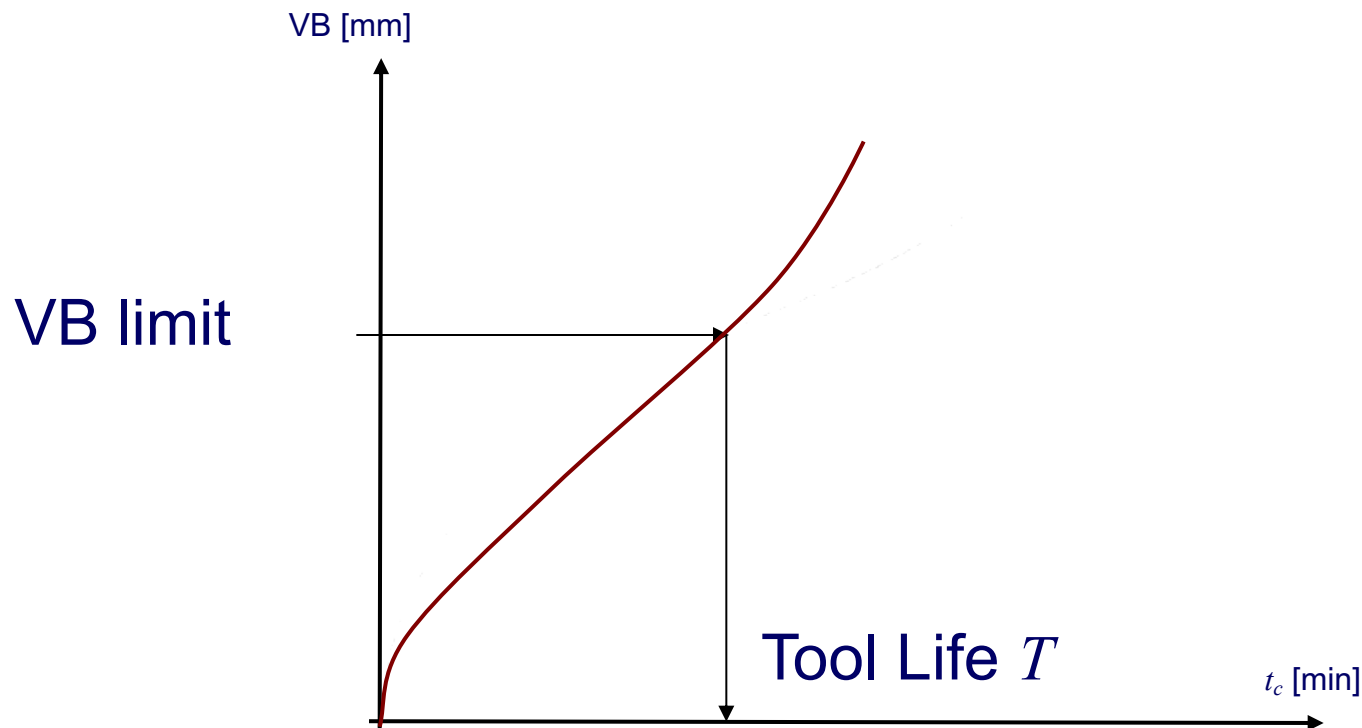
Effect of cutting speed on rake wear (KT)

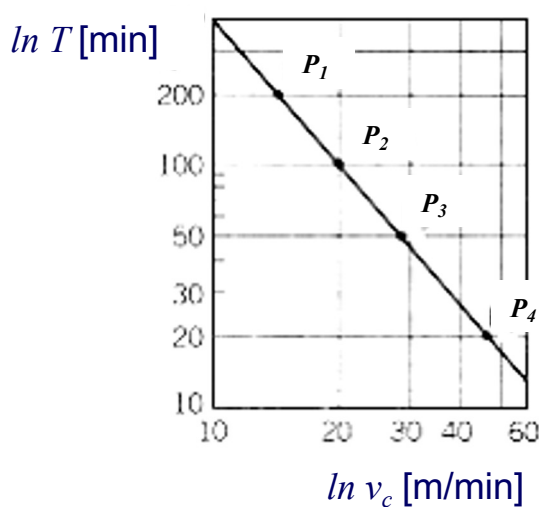
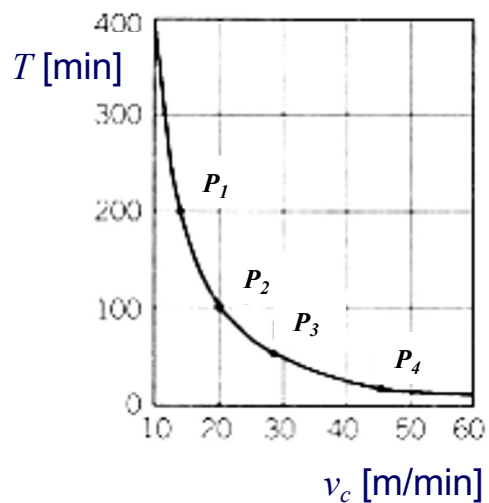
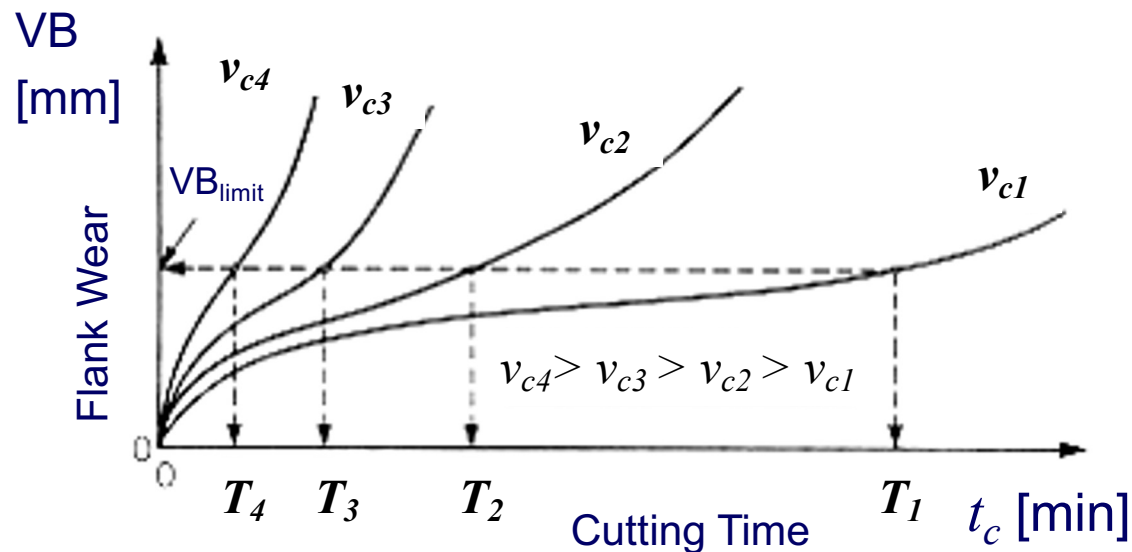




TOOL LIFE: FAILURE CRITERIA

We must define a ***tool failure criterion*** in terms of maximum wear level (for example $VB = 0,3 \text{ mm}$, $KT/KM = 0,1$) to determine the instant at which to change the tool.







Classical Taylor's law

$$v_c T^n = C$$

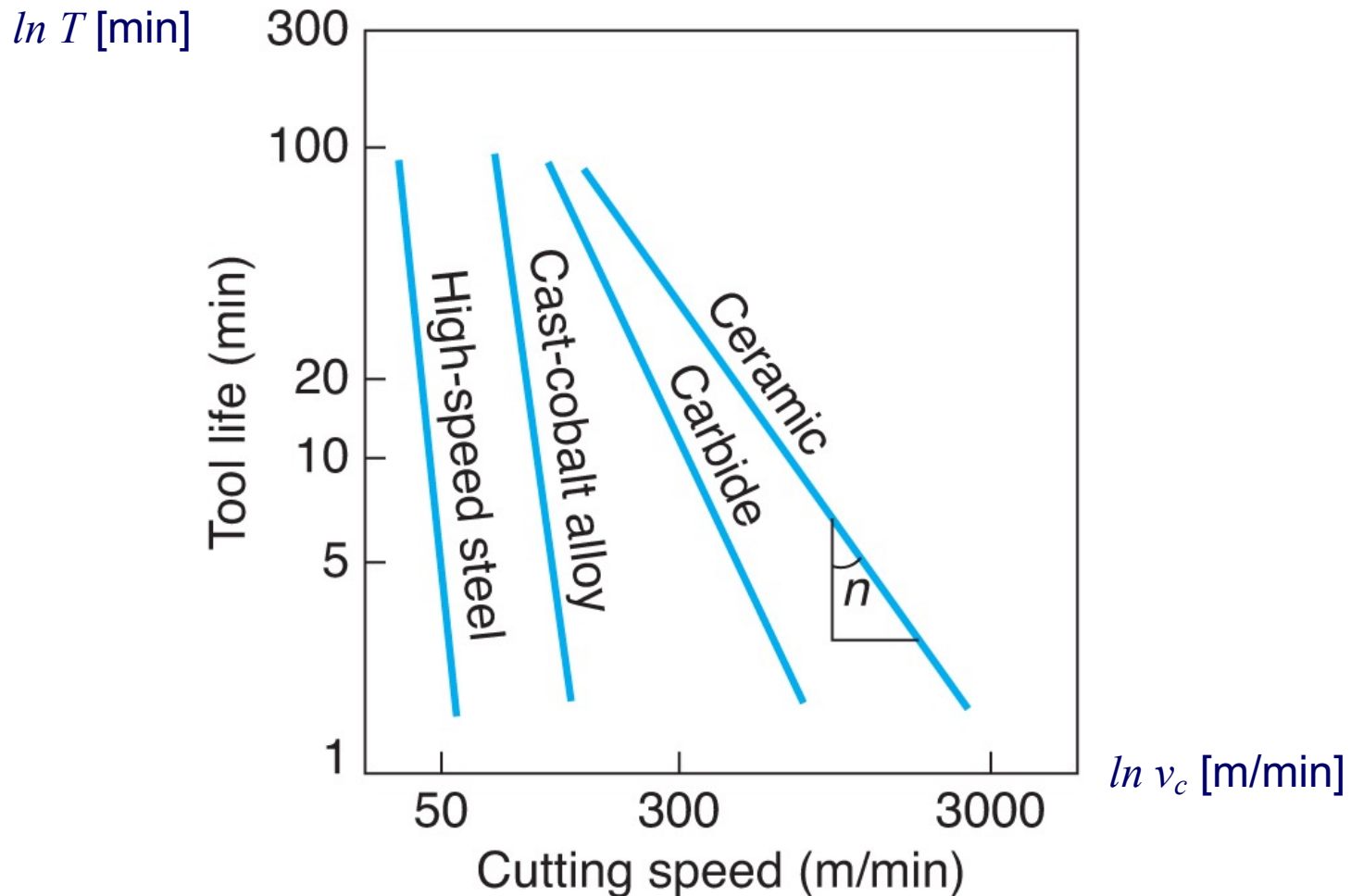
- v_c : cutting speed,
- T : tool duration or tool life,
- n : exponent depending mainly on the cutting tool material,
- C : parameter depending on the workpiece material and on the other machining parameters (feed and cutting depth).

Generalized Taylor's law

$$v_c T^n f^m a_p^q = C^*$$

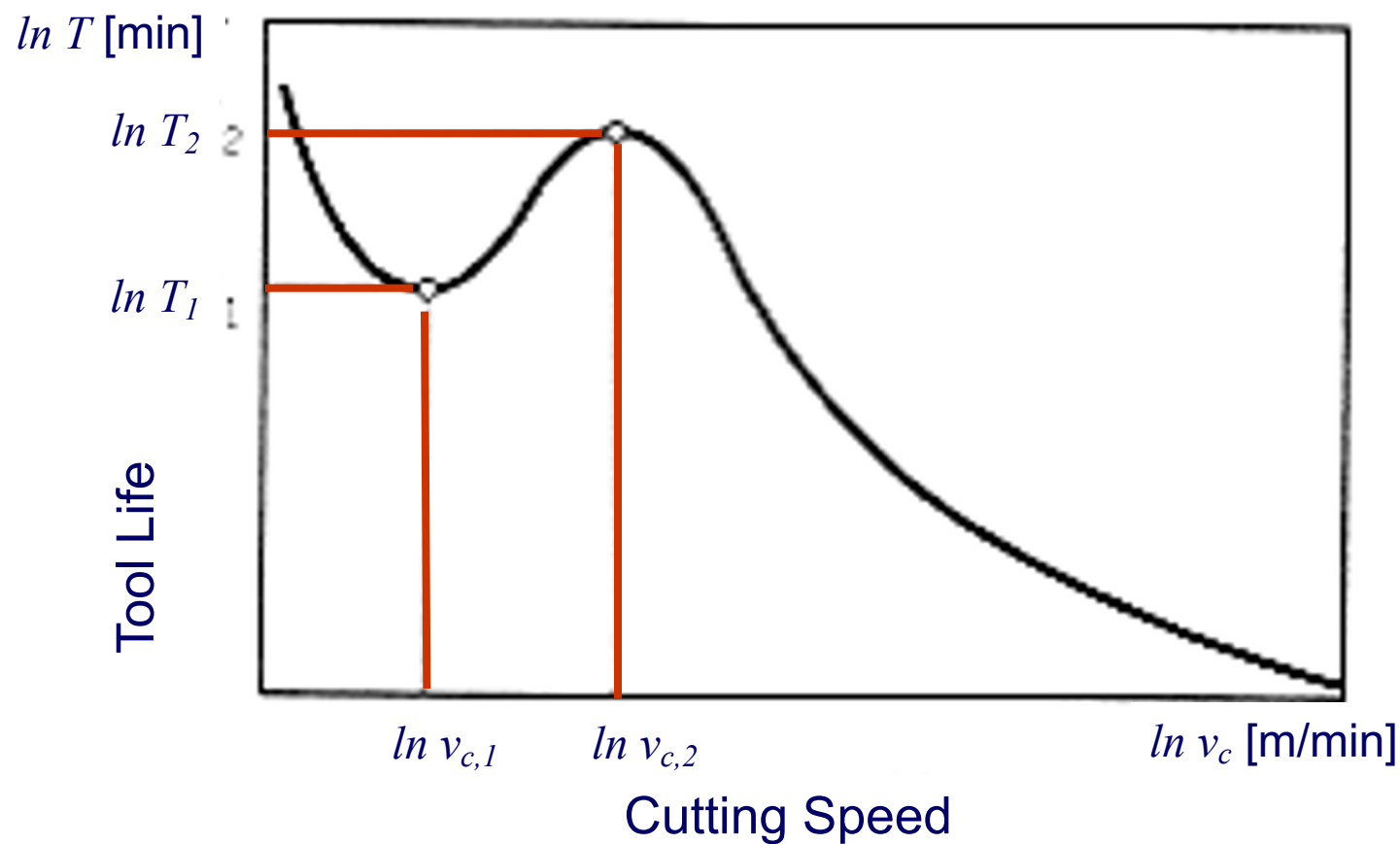


Tool life versus cutting speed for different tool materials.





Tool life versus cutting speed.





Main requirements for a cutting tool:

- Toughness
- Wear resistance
- Hardness at high temperature
- Heat conductivity
- Low thermal expansion coefficient
- Low friction coefficient
- Chemical inertia
- Limited cost



CUTTING TOOL MATERIALS

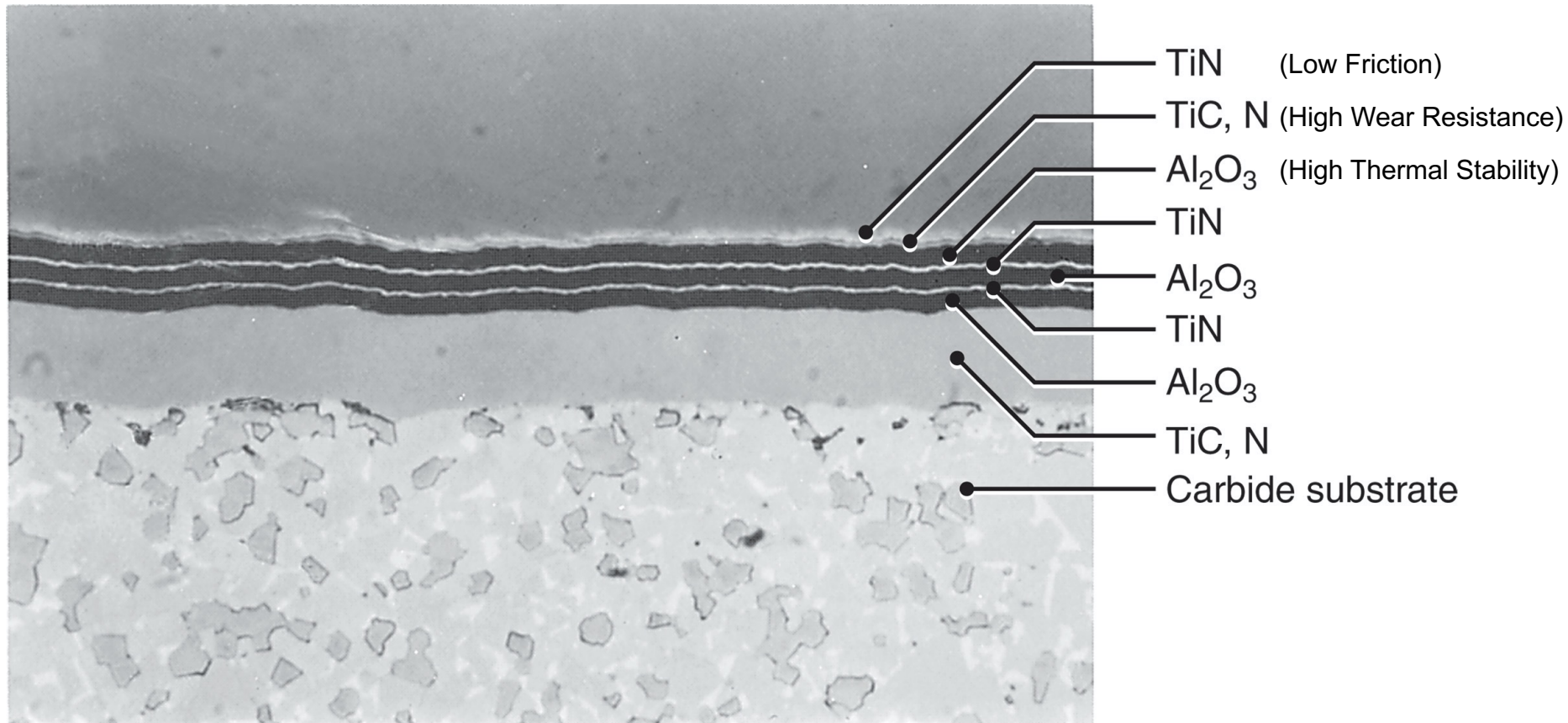
- Carbon steels
- High-speed steels
- Cast Alloys
- Carbides (coated or uncoated)
 - WC-Co,
 - WC-TiC-Co,
 - WC-TiC-TaC-Co
- Ceramics
 - Al_2O_3 , Si_3N_4
- Cermets (ceramic metals)
- CBN: Cubic Boron Nitride
- Diamond: natural, artificial, polycrystalline.

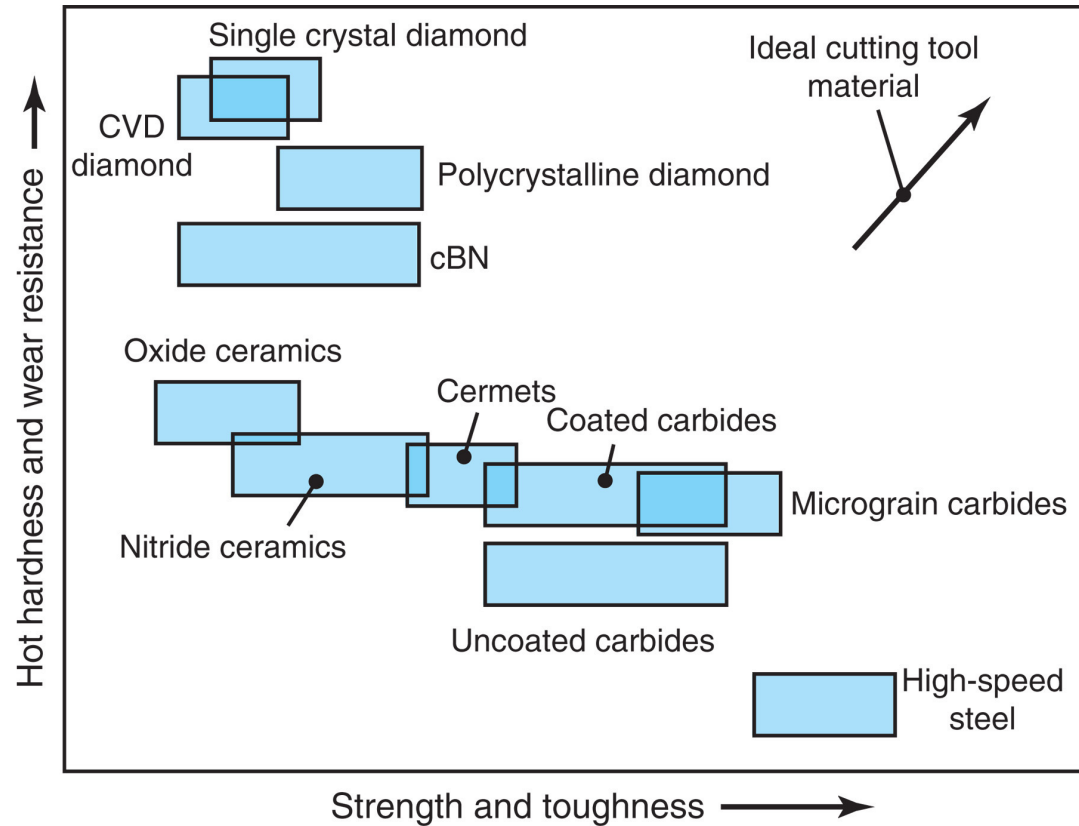
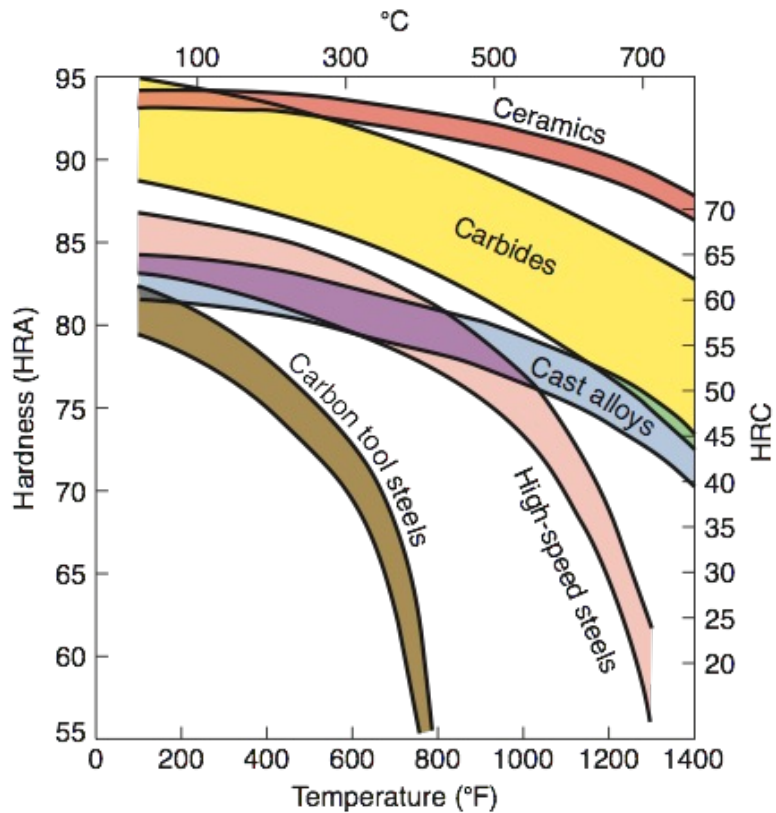
+ v_c
(with constant
Tool Life)





Multiphase coatings on a tungsten-carbide substrate; three alternating layers of aluminum oxide are separated by very thin layers of titanium nitride. Inserts with as many as 13 layers of coatings have been made. Coating thicknesses are typically in the range from 2 to 20 μm .





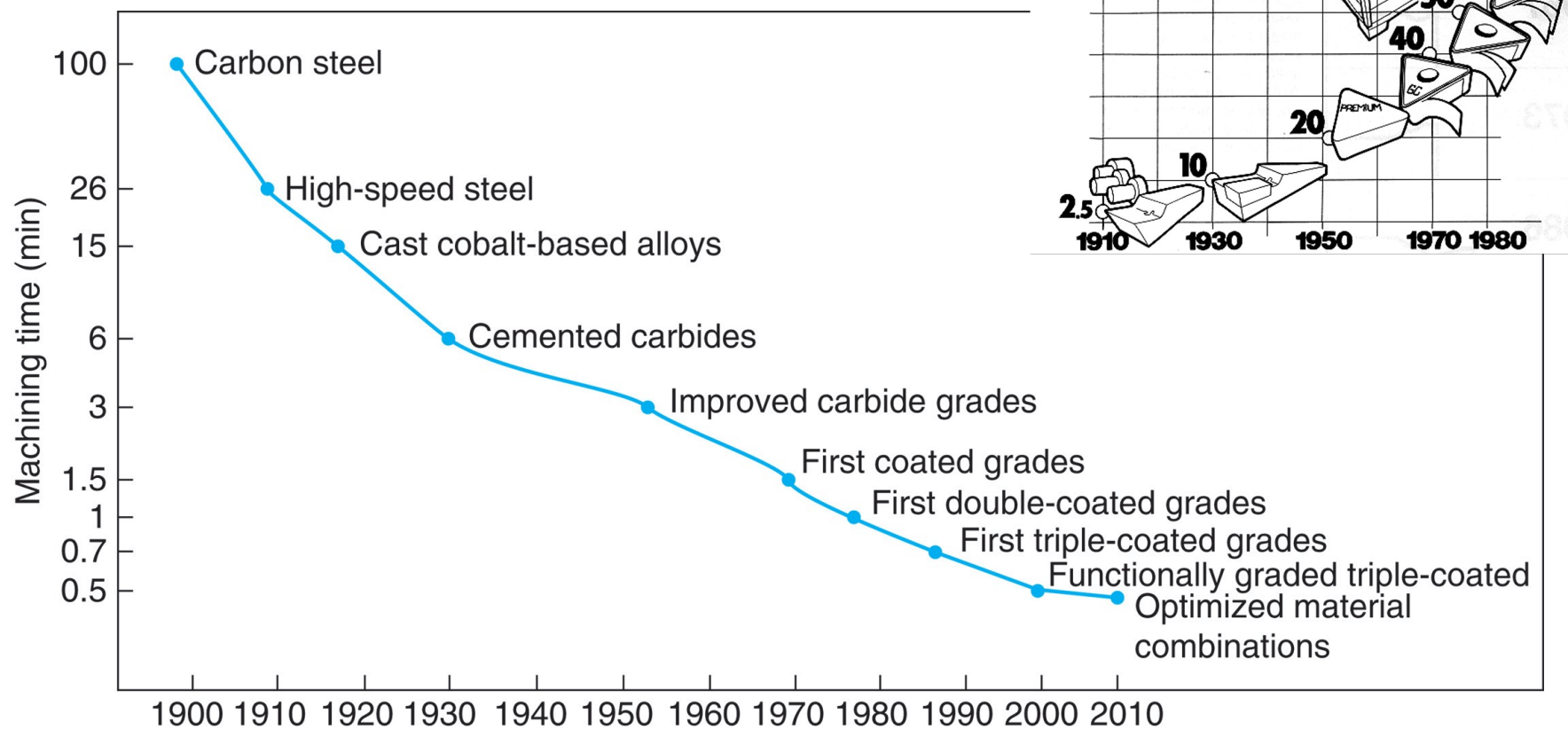


Less wear leads to:

- Increase of the production capacity
 - by increasing the removed chip volume,
 - by decreasing idle times for changing the worn tool.
- Decrease costs
 - by decreasing the tool cost per part.



Historical Evolution



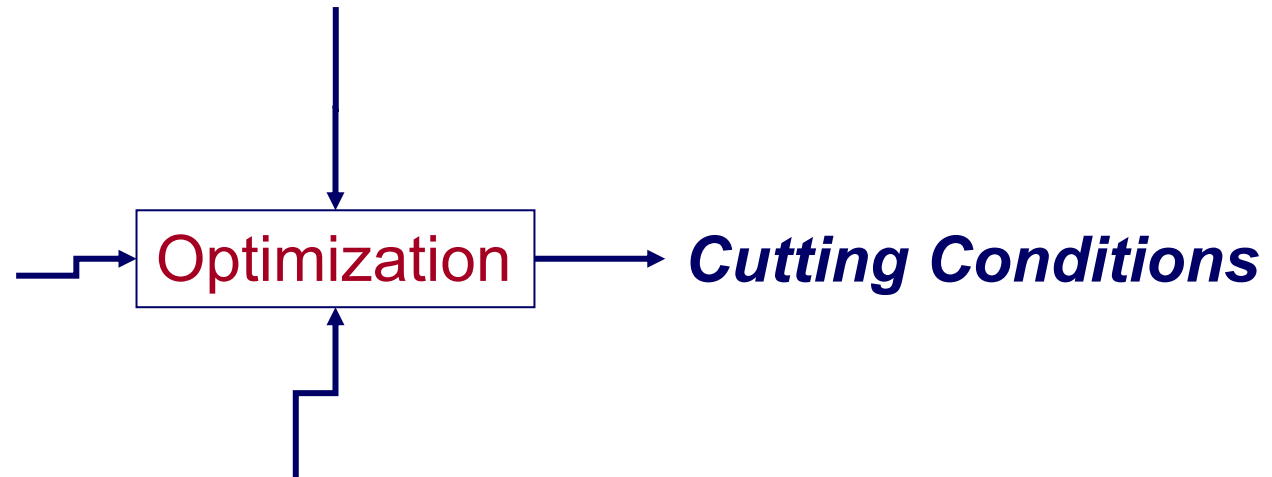


System

- decisional variables
- constants
- relationships
- constraints

Key Performance Indicators

(objective function)



Approach:

- analytical
- numerical
- experimental



A) *Production Time* [T_p]

$$T_p = T_h + T_c + \frac{T_t}{n_p} = T_h + \frac{V}{MRR} + \frac{V \cdot T_t}{MRR \cdot T} = T_h + \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right)$$

T : tool life [min.]

V : removed material volume [mm³]

$MRR = A_D \cdot v_c$: material removal rate [mm³ / min.]

T_h : fixed time [min.]

$T_c = \frac{V}{MRR}$: machining time [min.]

T_t : tool change time [min.]

$n_p = \frac{T}{T_c} = \frac{MRR \cdot T}{V}$: number of machined parts during tool life [min.]



B) *Production Cost* [C_p]

$$C_p = C_o T_p + \frac{C_t}{n_p} = C_o T_h + C_o \frac{V}{MRR} + C_o \frac{VT_t}{MRR \cdot T} + \frac{VC_t}{MRR \cdot T}$$

$$C_p = C_o T_h + C_o \frac{V}{MRR} \left[1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_o} \right) \right]$$

C_o : production resource cost per minute (machine tool and human operator) [€ /min.]

C_t : average cost of the cutting edge [€ /cutting edge]



Max Power $F_c v_c \leq \eta \cdot P_e$

Max workpiece bending $b_{wp} \leq \varepsilon_{wp}$

Max tool bending $b_{tool} \leq \varepsilon_{tool}$

Feed range $f_{min} \leq f \leq f_{max}$

Cutting speed range $v_{c,min} \leq v_c \leq v_{c,max}$

Roughness requirement $\frac{f^2}{32r} \leq R_a$



- Taylor's Law $v_c T^n = C$
- Cutting Force $F_c = k_c A_D$
- ...

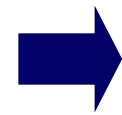


A) Production time [T_p]

B) Production cost [C_p]

$$T_p = T_h + \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right)$$

$$C_p = C_o T_h + C_o \frac{V}{MRR} \left[1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_o} \right) \right]$$



$$F = F_0 + \frac{\alpha}{MRR} \left(1 + \frac{\varphi}{T} \right)$$



If the optimization parameter is the cutting speed:

$$\begin{aligned} F &= F_0 + \frac{\alpha}{MRR} \left(1 + \frac{\varphi}{T} \right) \\ T &= T(v_c) \\ MRR &= A_D v_c \end{aligned} \quad \Rightarrow \quad F = F_0 + \frac{\alpha}{A_D v_c} \left(1 + \frac{\varphi}{T(v_c)} \right)$$
$$F = F_0 + \frac{\alpha'}{v_c} \left(1 + \frac{\varphi}{T(v_c)} \right)$$



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

$$\frac{dF}{dv_c} = 0$$

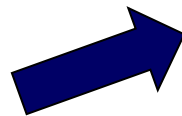
$$\frac{d}{dv_c} \left[F_0 + \frac{\alpha'}{v_c} \left(1 + \frac{\varphi}{T(v_c)} \right) \right] = 0$$

$$-\frac{\alpha'}{v_c^2} \left(1 + \frac{\varphi}{T} \right) - \frac{\alpha' \varphi}{v_c T^2} \frac{dT}{dv_c} = 0$$

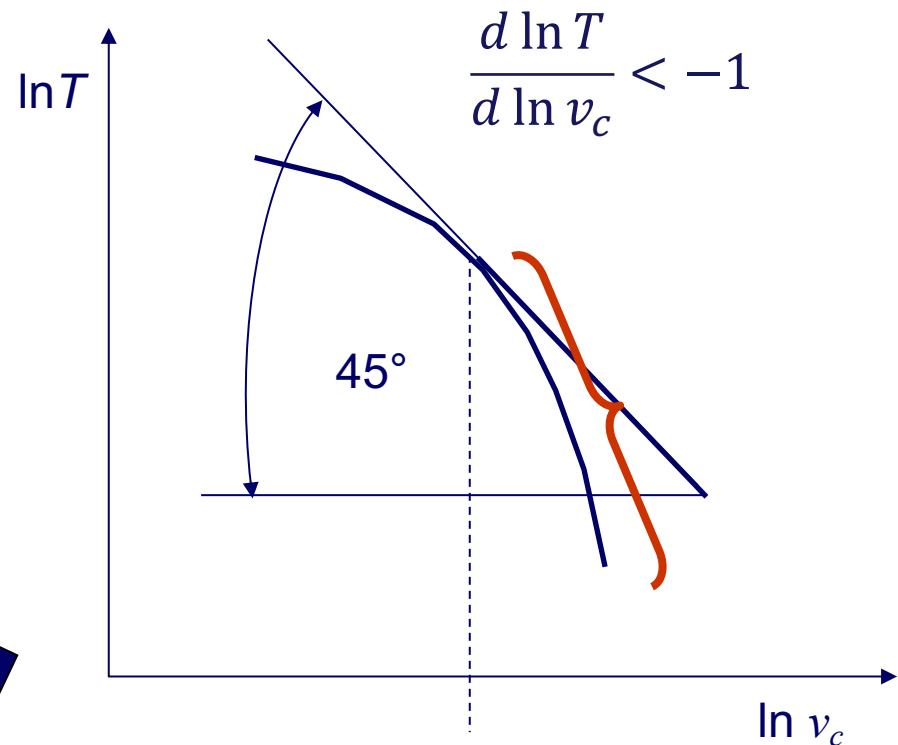
$$-\frac{\alpha'}{v_c^2} \left(1 + \frac{\varphi}{T} + \frac{\varphi v_c}{T^2} \frac{dT}{dv_c} \right) = 0$$

$$1 + \frac{\varphi}{T} + \frac{\varphi v_c}{T} \frac{dT}{dv_c} = 0$$

$$1 + \frac{\varphi}{T} \left(1 + \frac{d \ln T}{d \ln v_c} \right) = 0$$



An extreme point exists if:





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

Condition for the minimum

$$\frac{d^2 F}{dv_c^2} > 0$$

since $\frac{d \ln T}{d \ln v_c} < -1$

$$\frac{d^2(\ln T)}{d(\ln v_c)^2} < \frac{d(\ln T)}{d(\ln v_c)} \left(1 + \frac{d(\ln T)}{d(\ln v_c)} \right)$$

then $\frac{d^2(\ln T)}{d(\ln v_c)^2} < \delta$

< -1 < -1

< 0

With $\delta > 0$



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

Let's consider the Taylor's law: $T^n(v_c) = \frac{C}{v_c}$

In this case the value $\frac{d \ln T}{d \ln v_c}$ is: $n \ln T = \ln C - \ln v_c$
 $\ln T = \frac{1}{n} \ln C - \frac{1}{n} \ln v_c$
 $\frac{d \ln T}{d \ln v_c} = -\frac{1}{n}$

It follows that: $1 + \frac{\varphi}{T} \left(1 - \frac{1}{n}\right) = 0$

$$T_{opt} = \varphi \left(\frac{1}{n} - 1 \right) \rightarrow v_{c,opt} = \frac{C}{T_{opt}^n}$$



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

The optimal condition is structural, thus it holds both for T_p and C_p .

Since:

$$\varphi(T_p) = Tt < \varphi(C_p) = Tt + C_t/C_0$$

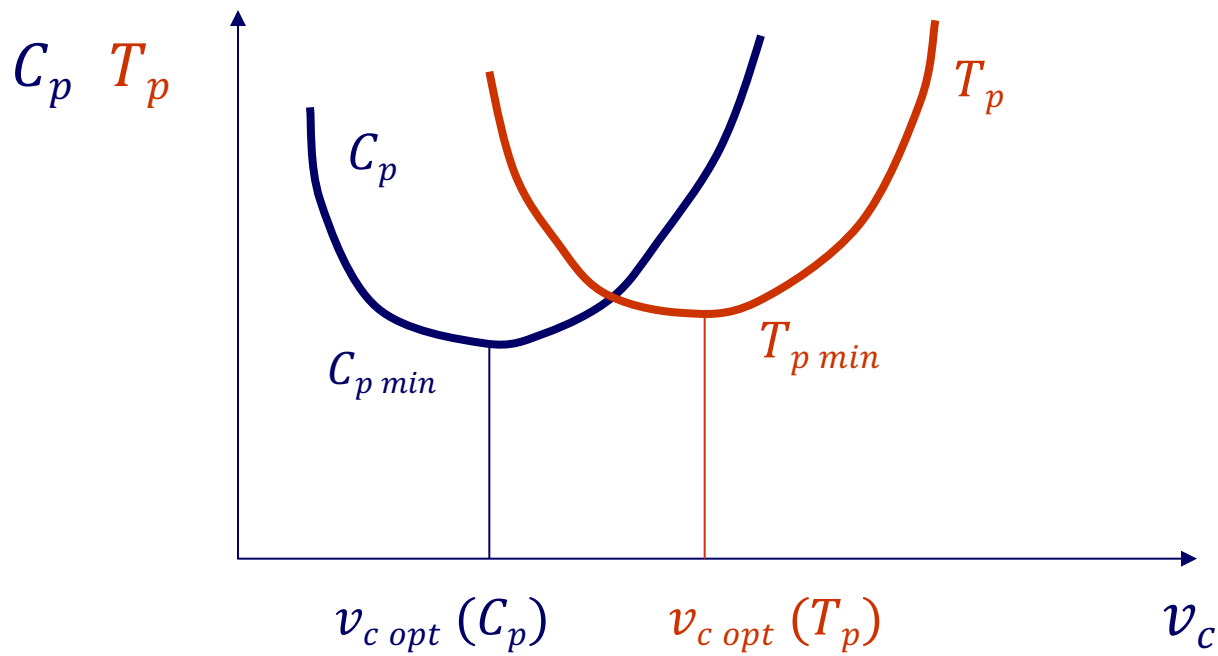
Then, it follows that:

$$T_{opt}(T_p) < T_{opt}(C_p)$$

$$v_{c\ opt}(T_p) > v_{c\ opt}(C_p)$$

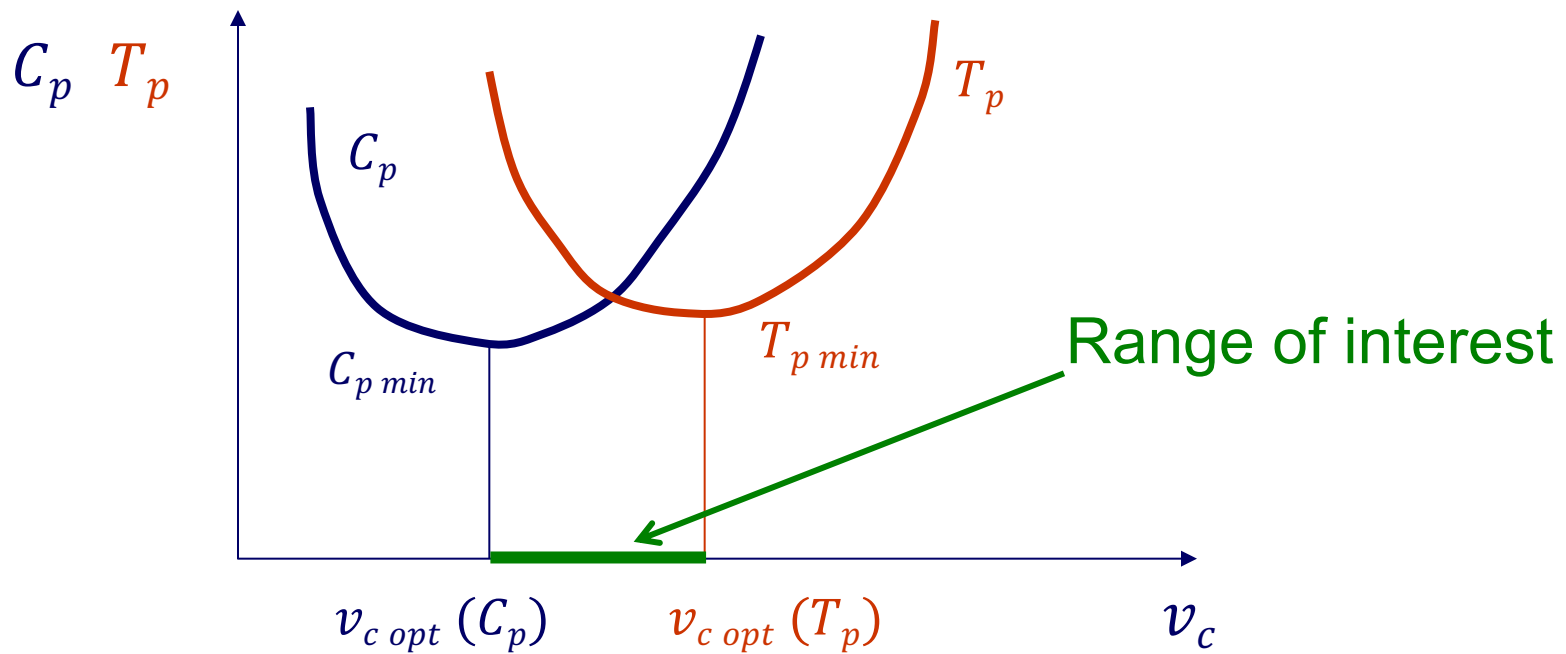


UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





C) *Profit Rate* [P_r]

$$P_r = \frac{R - C_p}{T_p}$$

where:

$R = \text{revenue}$

$$T_p = T_h + \frac{V}{MRR} \left(1 + \frac{T_t}{T} \right)$$

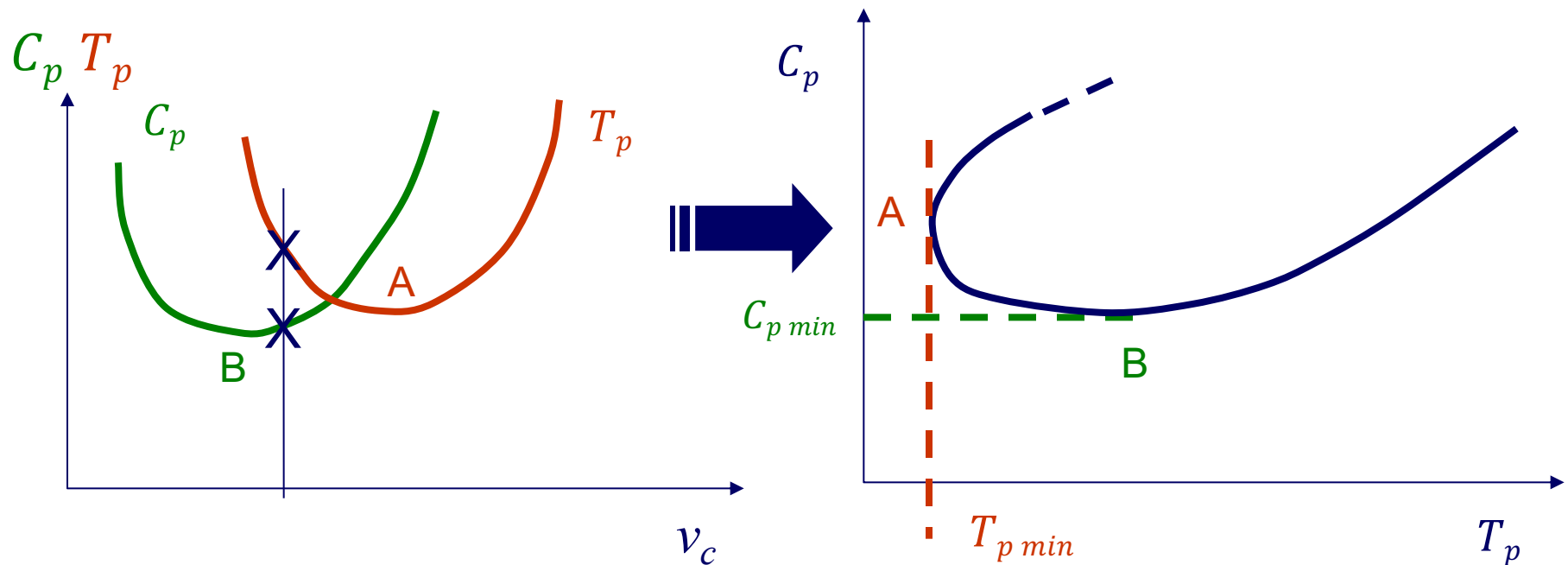
$$C_p = C_o T_h + C_o \frac{V}{MRR} \left[1 + \frac{1}{T} \left(T_t + \frac{C_t}{C_o} \right) \right]$$



UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

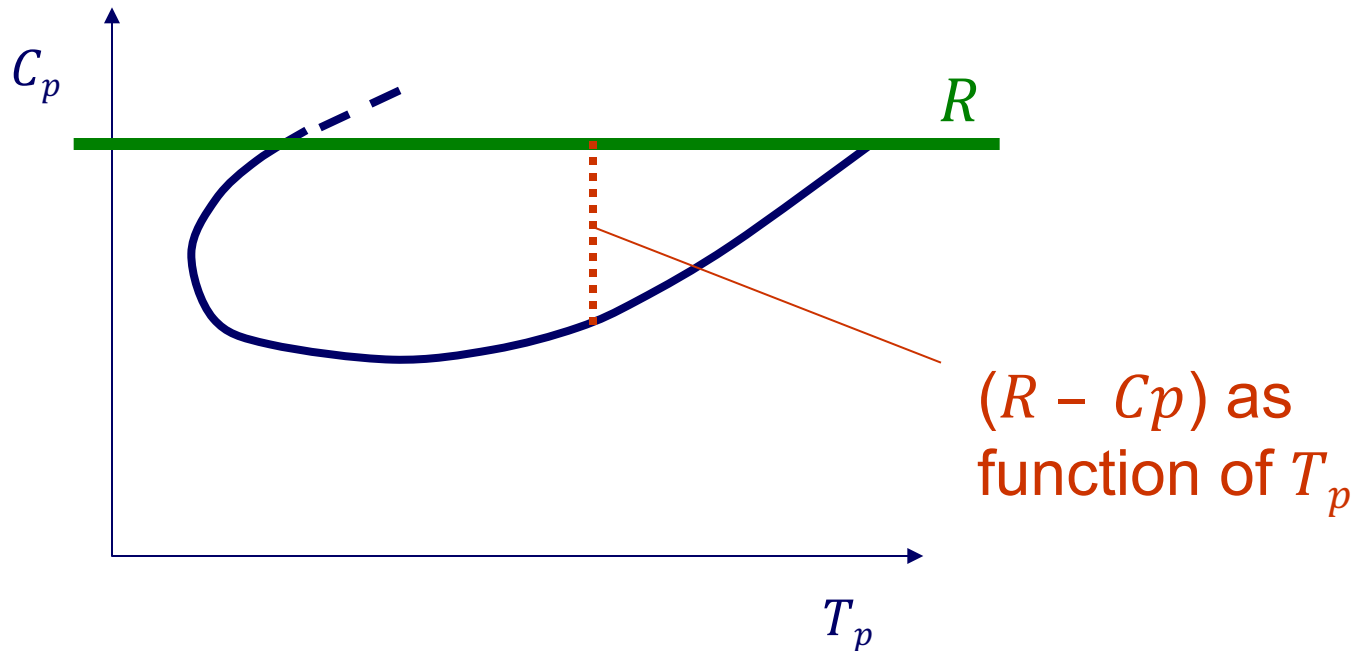
C) Profit Rate [P_r]

$$P_r = \frac{R - C_p}{T_p}$$



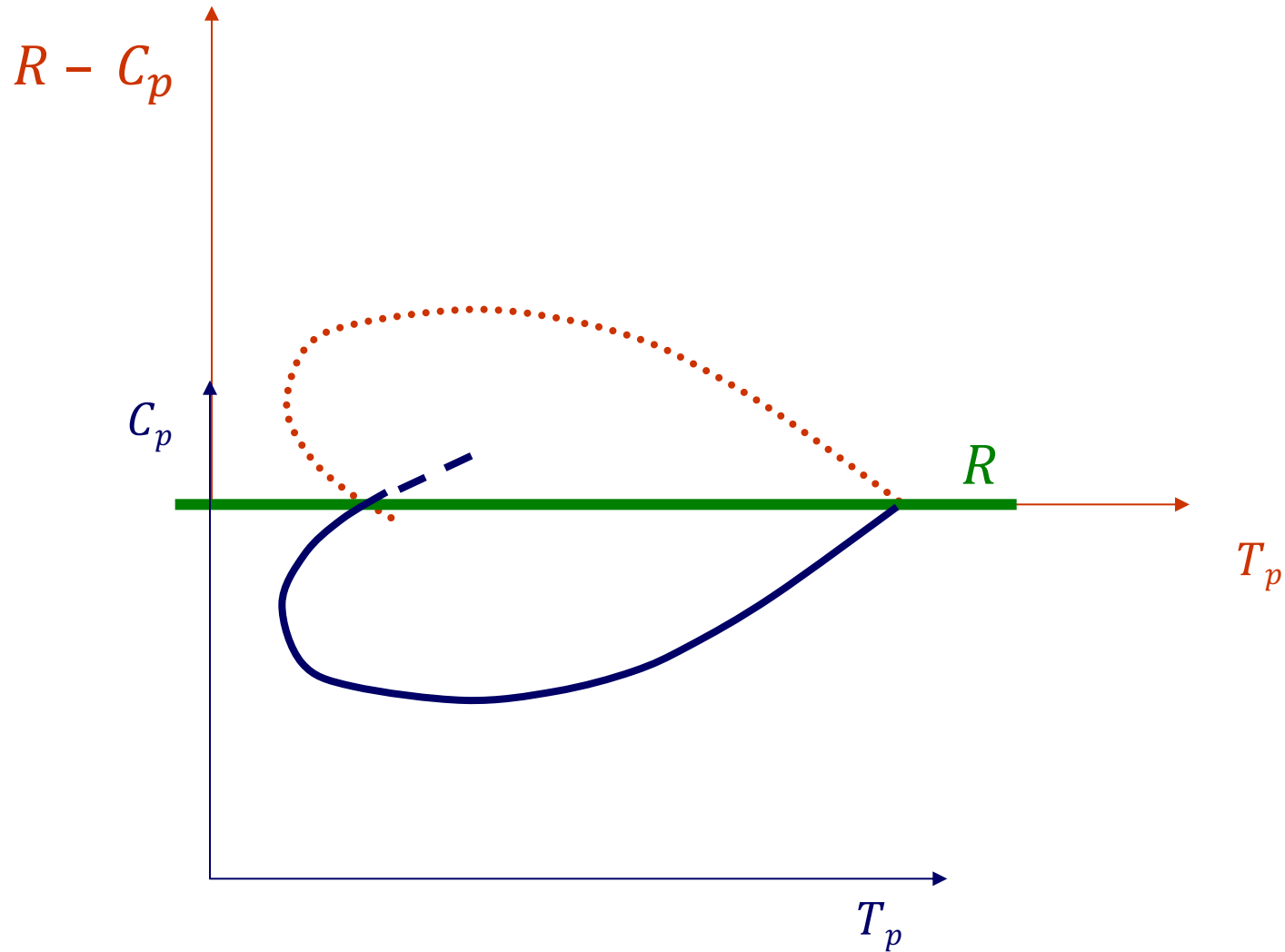


Assuming constant revenue R



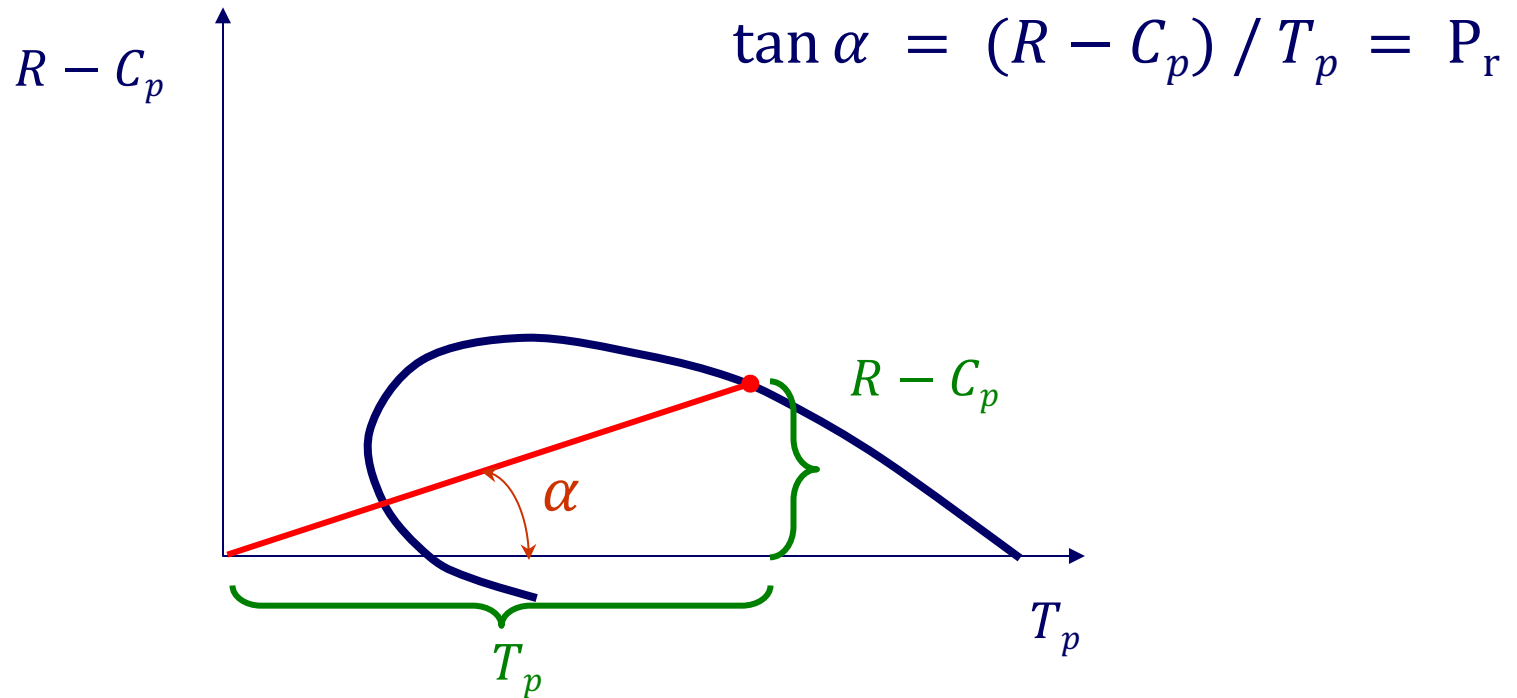


UNCONSTRAINED OPTIMIZATION: 1! VARIABLE



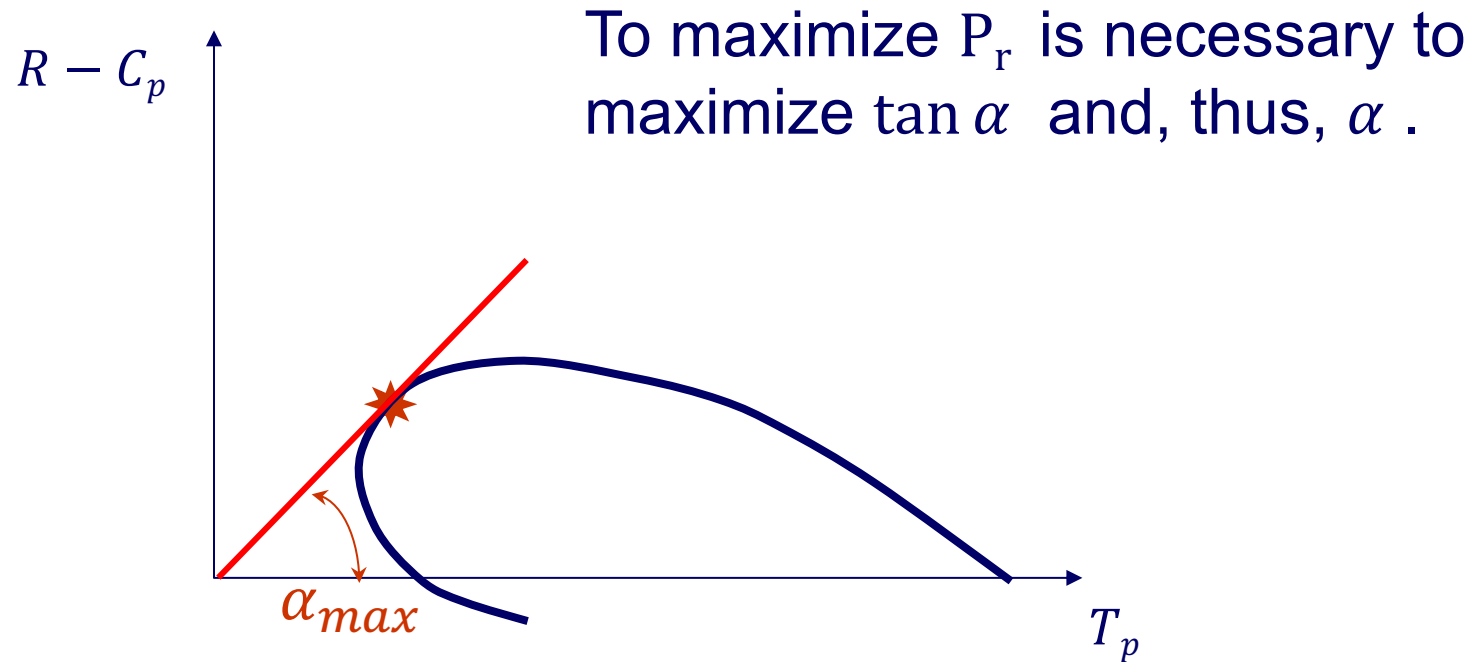


UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE





UNCONSTRAINED OPTIMIZATION: 1! VARIABLE

The P_r optimal condition is between the ones that minimize C_p and T_p (range of interest).

